

Experiment - 12

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(A)

Design a trigger which whenever there is an insertion on student table then, the currently inserted or deleted row should be printed as it as on the output console window.

Solution :

```
CREATE OR REPLACE FUNCTION FN_STUDENT_AUDIT()
RETURNS TRIGGER
LANGUAGE PLPGSQL
AS $$
BEGIN
    IF TG_OP = 'INSERT' THEN
        RAISE NOTICE 'INSERTED ROW -> ID: %, NAME: %, AGE: %, CLASS:
%',
            NEW.ID, NEW.NAME, NEW.AGE, NEW.CLASS;
        RETURN NEW;

    ELSIF TG_OP = 'DELETE' THEN
        RAISE NOTICE 'DELETED ROW -> ID: %, NAME: %, AGE: %, CLASS:
%',
            OLD.ID, OLD.NAME, OLD.AGE, OLD.CLASS;
        RETURN OLD;

    END IF;
```

```
    RETURN NULL;  
END;  
$$;
```

```
CREATE TRIGGER TRG_STUDENT_AUDIT  
AFTER INSERT OR DELETE  
ON STUDENT  
FOR EACH ROW  
EXECUTE FUNCTION FN_STUDENT_AUDIT();
```

```
INSERT INTO STUDENT(NAME, AGE, CLASS) VALUES('JOHN', 17, 11);
```

```
DELETE FROM STUDENT  
WHERE NAME='JACK';
```

(B) Design a PostgreSQL trigger that:

- Whenever a new employee is inserted in `tbl_employee`, a record should be added to `tbl_employee_audit` like: *"Employee name <emp_name> has been added at <current_time>"*
- Whenever an employee is deleted from `tbl_employee`, a record should be added to `tbl_employee_audit` like: *"Employee name <emp_name> has been deleted at <current_time>"*

Solution:

```
CREATE OR REPLACE FUNCTION AUDIT_EMPLOYEE_CHANGES()  
RETURNS TRIGGER  
LANGUAGE PLPGSQL  
AS $$  
BEGIN
```

```
IF TG_OP = 'INSERT' THEN
    INSERT INTO TBL_EMPLOYEE_AUDIT(message)
    VALUES ('Employee name ' || NEW.EMP_NAME || ' has been added at ' ||
NOW());
    RETURN NEW;

ELSIF TG_OP = 'DELETE' THEN
    INSERT INTO TBL_EMPLOYEE_AUDIT(message)
    VALUES ('Employee name ' || OLD.EMP_NAME || ' has been deleted at ' ||
NOW());
    RETURN OLD;
END IF;

RETURN NULL;
END;
$$;
```

```
CREATE TRIGGER TRG_EMPLOYEE_AUDIT
AFTER INSERT OR DELETE
ON TBL_EMPLOYEE
FOR EACH ROW
EXECUTE FUNCTION AUDIT_EMPLOYEE_CHANGES();

INSERT INTO TBL_EMPLOYEE(EMP_NAME, EMP_SALARY)
VALUES ('ALAN', 50000);

DELETE FROM TBL_EMPLOYEE
WHERE EMP_NAME = 'ALAN';
SELECT * FROM TBL_EMPLOYEE_AUDIT;
```